

Wind Turbine Design for Energy Extraction on Highways

EN 615: Wind Energy Conversion Systems

by

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Contents

1	Introduction and Problem Statement	2
2	Objective and Demand Estimation	2
2.1	Objective and Wind Turbine Selection	2
2.2	Demand Estimation	2
2.3	Wind Speed Analysis	3
3	Design Details	4
3.1	Sizing and Aerodynamic Analysis	4
3.1.1	Blade Profile	5
3.1.2	Aspect Ratio	6
3.1.3	No. of Blades	6
3.1.4	Overlap Ratio	7
3.1.5	End Plate Diameter	8
3.1.6	Blade Arc angle	8
3.1.7	Blade shape factor (p/q)	9
3.2	Design Specifications	10
3.3	Fabrication Approach and Assembly Steps	11
4	Performance Assessment	14
4.1	Observations from the Experiment	14
5	Experience	16
5.1	Work Allotment	17
6	Conclusion	17
	References	18

1 Introduction and Problem Statement

Our project tackles the challenge of harnessing wind power by designing a wind turbine specifically tailored for energy extraction along highways. Harnessing wind energy from vehicular movement on highways to generate sustainable electricity for highway lights. As Highways offer a unique opportunity due to their consistent wind flow, it makes them the ideal locations for wind turbine installation.

2 Objective and Demand Estimation

2.1 Objective and Wind Turbine Selection

We will be utilizing Vertical Axis Wind Turbines (VAWT) for the project, specifically a Savonius Wind Energy Turbine. We have chosen this type of wind-machine as it offers omnidirectional wind capture, minimal height requirements, and simplified maintenance, ideal for turbulent highway environments.

Our objective is to design a Savonius Wind Energy Turbine that can be installed on highway for energy extraction. We also aim to build a scale model of the Wind Energy Device and demonstrate the lighting of a 4W LED using our Savonius Wind Turbine

2.2 Demand Estimation

Highway street lighting is powered by a 30W LED using savonius wind turbine. Lower end of LED used on highway is around 30W, so we will be considering that as our end use power. The power of rotor that should be generated in order to achieve this task will be 30W which can be estimated using the $C_p - \lambda$ curve.

2.3 Wind Speed Analysis

Wind speed analysis is crucial for the design and performance assessment of our Savonius Wind Energy Turbine. To achieve this, we referred to literature for data on wind velocity and turbine speeds. Specifically, we utilized data from the study conducted by Santhakumar et al. in 2018 [4].

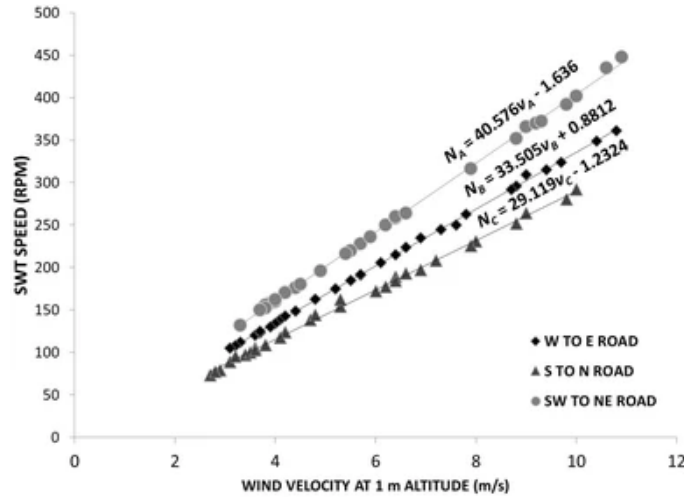


Figure 1: RPM v/s Wind Velocity graph for Savonius wind turbines

Their research involved regression analysis to establish the relationship between normal wind velocity at 1 meter altitude and the speeds of Savonius wind turbines (SWT). The graph depicted in above figure illustrates the correlation between RPM (rotations per minute) and wind velocity for Savonius wind turbines. Through curve fitting, the authors achieved high R-squared values exceeding 0.95, indicating a strong fit of the data.

By incorporating these findings into our project, we facilitate an accurate design specifications and performance predictions for our Savonius Wind Energy Turbine. Equations provided below can be used to extrapolate the speeds for unknown ranges.

$$N_A = 40.576v_A - 1.636; R^2 = 0.9991$$

$$N_B = 33.505v_B + 0.8812; R^2 = 0.9988$$

$$N_C = 29.119v_C - 1.2324; R^2 = 0.9994$$

3 Design Details

3.1 Sizing and Aerodynamic Analysis

Modelling of savonious wind turbine (SWT) to evaluate aerodynamic coefficients can be difficult for different designs, and the CFD simulations for multiple design stage was not feasible due to high computational time, thus we utilised data available from existing literatures to carry out conceptual design.

Reynold's No range was calculated using the formula $Re = \frac{\rho V D}{\mu}$, Where D is the span of the turbine. By limiting D to $0.5m$ due to size constraints, Reynolds Number (Re) for our setup will vary between $1.01 \times 10^5 - 1.69 \times 10^5$.

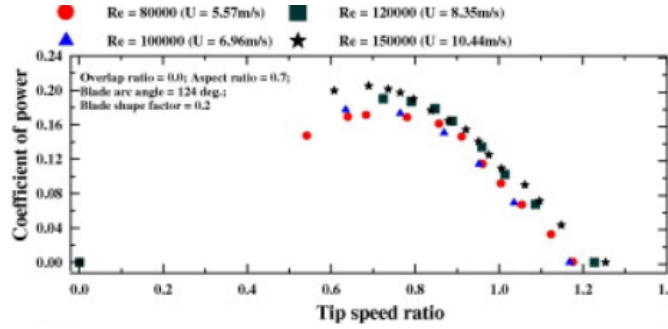


Figure 2: Cp vs tip speed ratio graph for SWT at different Reynold's number [1]

From the above figure 2, we conclude that for given Reynolds Number Re range (to maintain similar flow properties over the blades), at tip speed ratio(λ) 0.8, we get maximum C_p . Using this for further calulations. Considering design to be optimal around the velociy range 3m/s - 8m/s, the data related to similar SWT 1 shows that the RPM would vary between 100-250.

$$\lambda = u/V = \frac{\omega \times D}{2V} = \frac{RPM \times \frac{2\pi}{60} \times D}{2V} \quad (1)$$

From this data, we obtain $RPM/V = 33$ from the slope of RPM vs V graph. Using equation 1, assuming $V_{avg} = 5\text{m/s}$, and $\lambda = 0.8$, we obtain $D = 0.46\text{m}$.

Savonius Rotor Efficiency depends majorly on the following design parameters as follows, we have done literature survey to find the optimal design parameters for maximum Coefficient of Power (C_p)

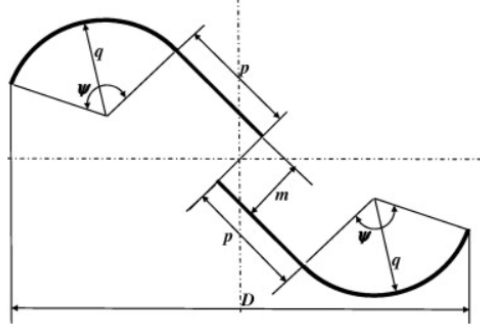


Figure 3: SWT design parameters[1]

3.1.1 Blade Profile

For Savonius Rotors, the Blade Profile plays an important role in terms of performance and efficiency, we can either use a semi-circular blade profile that is used for classical, or we can either use an airfoil-based, or shape optimized blade design.

Upon literature survey, we found this paper, which had compared the performance of different savonius blade designs as follows [2]

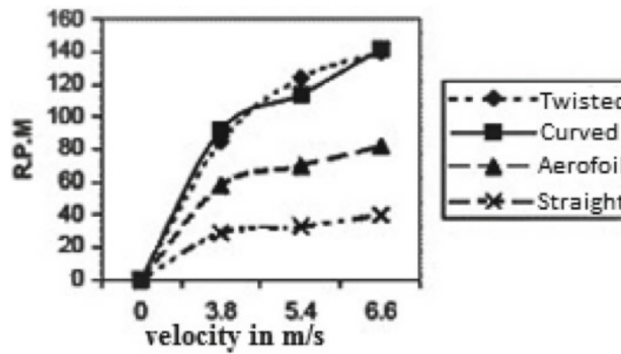


Figure 4: Performance of Different Blade Profiles [2]

The curved blade design has proven to be the most efficient among various blade designs such as straight and aerofoil, across a wide range of wind velocities as shown in the figure above.

3.1.2 Aspect Ratio

The aspect ratio, defined as the ratio of the rotor's height to its diameter (H/D), is a parameter influencing the performance of Savonius rotors. Through a literature review, we have identified a paper containing experimental data from a study conducted on rotors with an overlap ratio of zero and varying aspect ratios (H/D) of 0.6, 0.7, 0.77, and 1.0 [1]. The performance graph from the paper are as follows

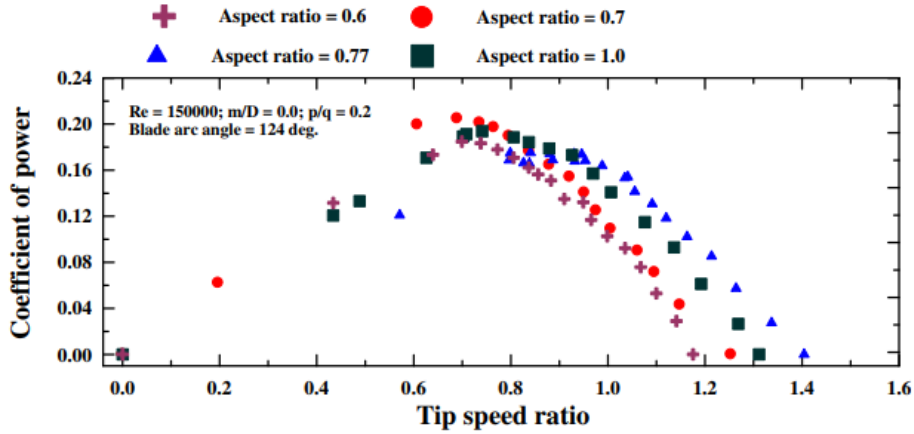


Figure 5: Effect of Aspect Ratio on the Coefficient of Power (C_p) [1]

The experimental results indicate that the rotor with an aspect ratio of 0.77 exhibits the highest coefficient of power at the tip speeds of $\lambda = 0.8 - 1$, hence we fix the aspect ratio at 0.77.

3.1.3 No. of Blades

The number of blades in a Savonius rotor significantly influences its performance. According to findings from a comparative study [3], the power coefficients for a

two-blade rotor surpass those of both three and four-blade rotors as shown in the figure below.

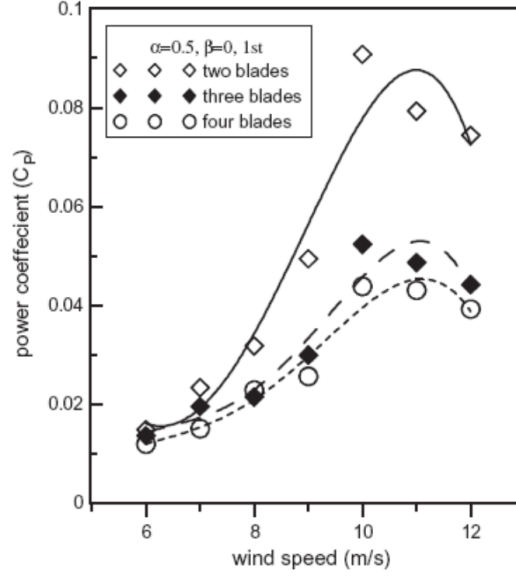


Figure 6: Coefficient of Power v/s No. of Blades [3]

This superiority is attributed to the higher net drag force experienced by the two-blade rotor compared to its counterparts. Thus, for our Savonius rotor design, we opt for a two-blade configuration to maximize efficiency based on these insights.

3.1.4 Overlap Ratio

The overlap ratio, denoted as the ratio of the rotor's overlapping area to its total rotor area, impacts the performance of Savonius rotors. In the study conducted by S. B. Kedare et al. [1], Savonius rotors with a zero overlap ratio demonstrate higher coefficients of power as shown in the below figure.

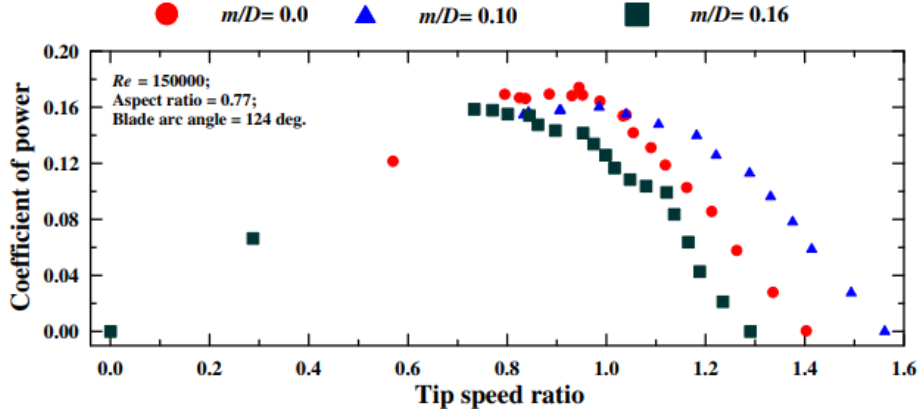


Figure 7: Coefficient of Power for different Overlap Ratios [1]

However, considering the incorporation of a shaft into the design and the negligible difference in power coefficient for overlap ratios of 0 and 0.1 at tip speed ratios above 0.8, we select an overlap ratio of 0.1 suitable for our Savonius rotor design.

3.1.5 End Plate Diameter

As we are following the study conducted by S. B. Kedare et al. [1], we have decided to incorporate the end plate parameter as $D_0/D = 1.1$ as mentioned in the study.

3.1.6 Blade Arc angle

The blade arc angle, which determines the curvature of the Savonius rotor blades, is a critical parameter affecting its performance. Research conducted by S. B. Kedare et al. [1] investigated the influence of blade arc angles of 110°, 124°, 135°, and 150° as shown in the figure below.

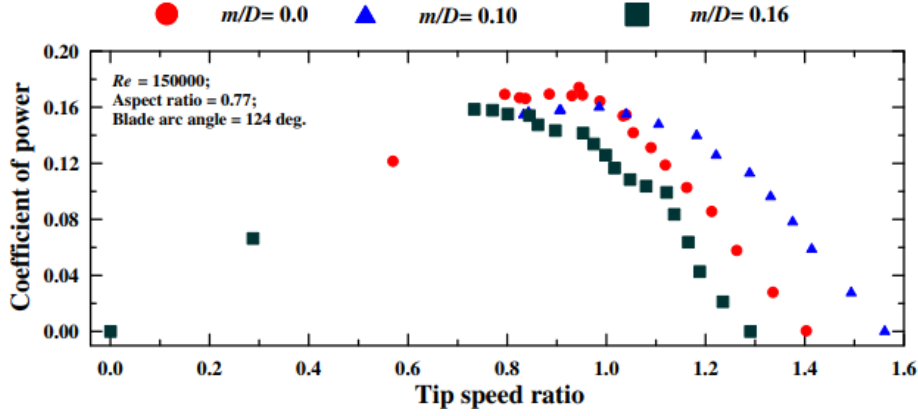


Figure 8: Coefficient of Power for different Blade Arc Angles [1]

At our anticipated tip speed ratios of 0.8 – 1, the rotor with a blade arc angle of 124° exhibits better performance. Hence, we adopt this angle for our Savonius rotor design.

3.1.7 Blade shape factor (p/q)

The blade shape factor (p/q) is a parameter influencing the performance of Savonius rotors. Research conducted by S. B. Kedare et al. [1] investigated the effects of different blade shape factors (0.2, 0.4, and 0.6) on the rotor's efficiency.

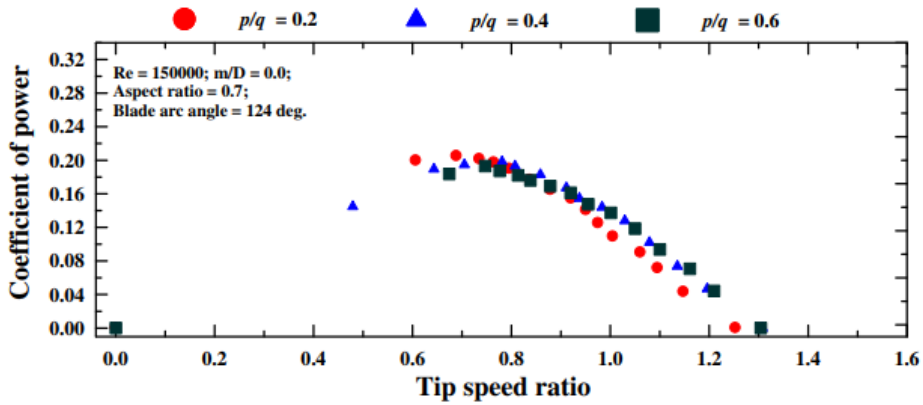


Figure 9: Coefficient of Power for different Blade Shape Factors [1]

The above figure depicts the variations in coefficient of power for these blade shape factors. We observe that the blade shape factor of 0.4 gives us the highest

power. Hence we fix this value for our design.

3.2 Design Specifications

Final design parameters determined based on above calculations and observation are as follows:

- **Diameter of Rotor** = Assuming $V_{avg} = 5m/s$, and $\lambda = 0.8$, and using Equation-1, we obtain $D = 0.46m$
- **Height of the Rotor** Using $AR = H/D$, from the study in Section 3.1.2, we obtain $H = AR \times D = 0.35m$
- **No of blades** = 2 (based on the study in sec 3.1.3)
- **Overlap (m)** , Using overlap ratio as 0.1 as per our design configuration, $m = D \times \text{Overlap Ratio} = 0.046m$
- **End Plate Diameter D_0** Taken as $1.1 \times D = 0.506m$
- **Blade Arc Angle** : 124 degrees based on design specification study conducted above.
- **p** : From the geometry shown in fig 3, relation between p, q, ψ and D can be written as

$$p \cos(\psi/2) + 2q \cos(90 - \psi/2) = D/2 \quad (2)$$

Also, from the blade shape factor, $p/q = 0.4$. Solving Equation, we get $p = 0.0449m$

- **q** : Using Equation-2 and blades shape factor value, we obtain $q = 0.1122m$

Below is the configuration image of the Setup

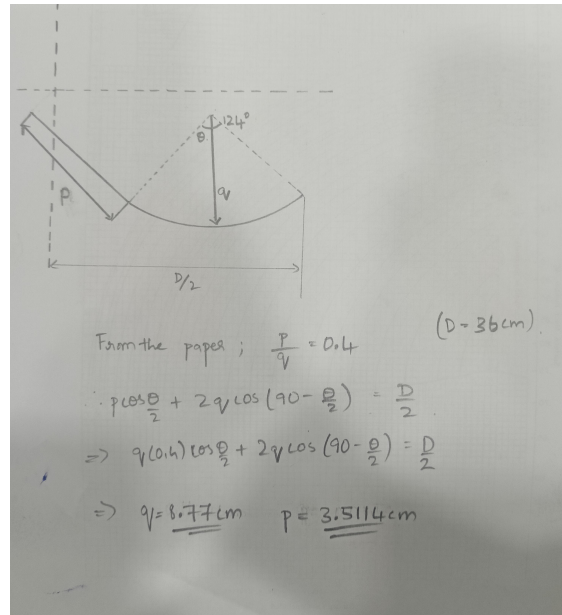


Figure 10: Design Specifications

3.3 Fabrication Approach and Assembly Steps

We had used commonly handy and available materials for fabrication of the wind turbine. We have used the following materials for the fabrication part:

1. Cardboard (Book Back and Front Covers)
2. Fevikwik, Fevicol, Glue gun as adhesive materials
3. Steel rod as rotating shaft
4. Depron and cardboard as frame work materials
5. DC Generator - 6-12 V for converting the mechanical motion of shaft into electrical work
6. LEDs - 3.4 V, 0.1 W as a load
7. Wire for connecting the load and the generator

Below are the steps that we used for creating the scale model

1. Material Assembly:

- Created dimensions on depron and cut it accordingly for bottom and top end plates.
- Utilized depron for both end plates.

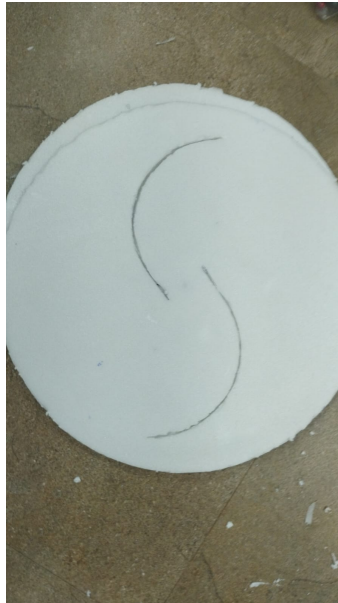


Figure 11: Depron Sheet Cutouts for End-Plates

2. Installation of Shaft Attachment:

- Took a small disc with a small orifice onto the bottom End-Plate.
- Ensured the orifice accommodated the DC Generator shaft snugly, facilitating the transmission of rotating power from the rotor to the shaft.

3. Submersion and Base Support:

- Positioned the generator atop a small cardboard box.
- Employed the small cardboard box as a supportive base for the wind turbine assembly.



Figure 12: Cardboard Base and Generator Housing

4. Covering and Shaft Integration:

- Enclosed the wind turbine within a complete big cardboard box.
- Inserted the wind turbine shaft, equipped with a bearing, through the top of the cardboard box.
- Utilized the bearing to facilitate shaft rotation and enhance overall stability of the setup.



Figure 13: Bearing and Shaft Integration

5. Purpose of Cardboard Cover:

- Utilized cardboard cover to provide structural support to the wind turbine assembly.
- Minimized oscillatory motions of the wind turbine, thereby ensuring stability and stillness of the setup.

We finally have out Savonius Wind Machine made from household materials as shown in the figure below.



Figure 14: Final Model of the Wind Machine

4 Performance Assessment

4.1 Observations from the Experiment

Upon assembling the apparatus, we proceeded to integrate the LED and resistor onto the breadboard, connecting them to the output of the generator. To simulate wind conditions, we used a Hair Dryer to replicate airflow. Initially, the LED Bulb did not turn on, leading us to investigate on the power generated by the DC generator. Using a Multimeter, we measured the Voltage and Current while the

rotor was in motion. We also measured the RPM using frame analysis from the video we recorded. Below are the results yielding from the experiments.

Dryer Mode	Wind Speed	RPM	Voltage V	Current I	P_{actual}	TSR λ	P_{rotor}	C_p
Low	4	100	0.4	0.5	0.2	0.8114	0.3	0.1446
Medium	8	140	0.5	0.6	0.3	0.5679	0.45	0.02713
High	10	180	0.6	0.8	0.5	0.5842	0.75	0.02315

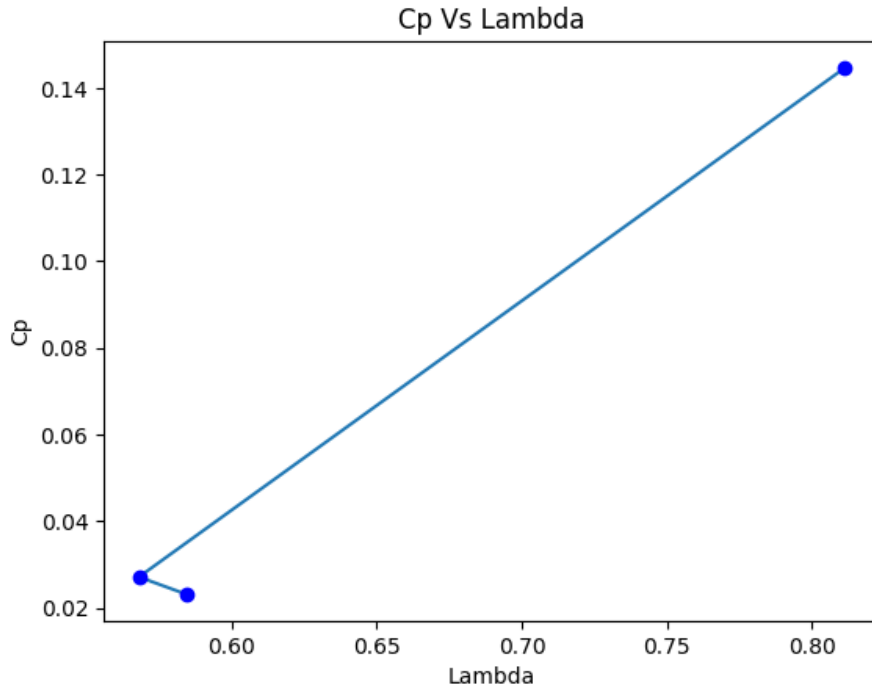


Figure 15: $C_p - \lambda$ Curve for Experiment

We observe that the plots follow the general trend of $C_p - \lambda$ curves. These voltage readings fell short of the typical requirement for LED operation, typically around 1.7-1.8V, and consequently, the generated power proved insufficient to light the LED bulb.

The video recording of us trying to directly light the LED is linked here

We also observe that the RPM that we expected the rotor to run at higher wind speeds is lower than the RPM it is running under a hair blower. This discrepancy may be attributed to various factors, including component rigidity and rotational resistance caused by cardboard components and bearings. Consequently, we fell short of achieving design conditions. Nonetheless, despite these challenges, our wind-machine yielded a commendable power output.

5 Experience

Through this project, we learned about various types of wind machines through research and study and realized the disparity between theoretical knowledge and practical implementation. We also delved into multiple research papers, gaining insights into the design of wind energy technologies.

- Customized Blade Design: Modified blade design to optimize power generation, specifically targeting highway wind speeds.
- Identified stability issues due to the rotor's large surface area compared to the motor's disc.
- Observed rotor wobbling along the center axis during rotation.
- Implemented a box stand around the turbine and utilized ball bearings to ensure consistent shaft rotation.
- Acknowledged unforeseen losses associated with ball bearing utilization, impacting achieved RPM.
- Collaborated effectively with team members and learning from them.
- Recognized the complexities of real-world engineering and the importance of adaptability and problem-solving skills.

Overall, the project provided a brilliant learning experience, enabling us to navigate the complexities of wind turbine technology.

5.1 Work Allotment

We were 4 members in the team and the following was the work allotment amongst the team members.

- **Ammar Barbhaiwala (200010010):** Worked on the design aspect of the wind machine and conducted literature survey to select the optimal design parameters and performed design calculations. Also worked on creating the physical model of the wind-machine along with other team members
- **Shambhu Kumar (200010072):** Worked on the design aspect of the wind machine and performed design calculations for specifications of the scaled model and conducted literature survey. Also worked on creating the physical model of the wind-machine along with other team members.
- **Aditya Vibhute (21D170003):** Worked on the Demand Estimation of Wind on Highway and Wind Analysis. Also worked on acquiring materials for the project and fabrication of the model along with other team members.
- **Manish H T (21D1700024):** Worked on the Demand Estimation of Wind on Highway and Wind Analysis. Also worked on acquiring materials for the project and fabrication of the model along with other team members.

All team members have contributed equally to the project and helped in the successful fabrication of the model.

6 Conclusion

The wind energy project undertaken by our team aimed to harness wind power along highways to provide sustainable electricity for highway lighting. Through design calculations, and fabrication efforts, we created a Savonius Wind Energy Turbine tailored for this purpose. Despite encountering challenges during the experimental phase, such as discrepancies between theoretical predictions and practical results,

we gained insights into wind turbine technology.

Based on the design, fabrication and experiment performed using the model, we can conclude the wind turbine designed by us which had designed to light an LED bulb of 1W, the turbine was only able to produce half the power required for lighting of the LED. This is due to the material used for fabrication, we had chosen book cardboard for the blades, instead if we had chosen another material like PVC pipe, we could have achieved what we had aimed for. Also, through literature survey and calculations, we concluded the number of blades to be 2, this gives us higher power at the rotor and low and high wind speeds. The wind blades were not cut properly at designed angles, and this caused some errors in the calculations that we did, thus we got approximately 50% of the power output. We can improve on this model further in terms of fabrication to extract more power from the wind.

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